
Influence of topical administration of n-3 and n-6 essential and n-9 nonessential fatty acids on the healing of cutaneous wounds

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Injury triggers a series of physiological events at the wound site. These include an inflammatory response that is established shortly after the injury, which is then followed by an intense formation of tissue over a period of days. Poly- and monounsaturated fatty acids exert major functions on the inflammatory responses, either in the form of phospholipids anchored in the cell membrane or as soluble lipoic mediators. We present evidence that linolenic (n-3), linoleic (n-6), and oleic (n-9) fatty acids can modulate the closure of surgically induced skin wounds. We found that n-9 fatty acids induced faster wound closure when compared to n-3, n-6, and control. In addition, n-9 fatty acids strongly inhibited the production of nitric oxide at the wound site. A mild improvement on wound closure was observed in the n-6 fatty acid-treated animals concurrent with a peak in nitric oxide production at 48 hours postsurgery. N-3 fatty acid treatment significantly delayed wound closure. Furthermore, we showed that n-3 fatty acid induced a peak in nitric oxide at 3 hours postsurgery and an intense deposition of extracellular matrix after 5 days of treatment. Thus, our results suggest a relevant role and potential therapeutic implication for fatty acids on skin wound healing. (**WOUND REP REG 2004;12:235-243**)

Wound healing involves a sequence of molecular and cellular events including inflammation, cell migration, angiogenesis, extracellular matrix synthesis, and re-epithelialization.¹ A number of biological mediators are required to control these different processes. Nitric oxide (NO) plays an important role in wound healing of the skin. It influences the functions of macrophages, fibroblasts, and keratinocytes during the healing process, contributing to re-epithelialization.² In animals lacking

CTF	Connective tissue fiber
iNOS	Inducible NO synthase
NK	Natural killer
NO	Nitric oxide
PBS	Phosphate buffered saline solution
PUFA	Polyunsaturated fatty acid

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the inducible form of nitric oxide synthase (iNOS) there is a delay in healing of skin wounds.³ Moreover, inhibition of NO synthesis causes fibroblasts and inflammatory cells to release other mediators that will decrease collagen deposition in the wound site.⁴

Injury activates a cascade of local and systemic immune responses,⁵ and the process of wound healing begins with an inflammatory reaction that requires complex interactions between a variety of cell types.⁶ Polymorphonuclear leukocytes and lymphocytes are attracted by soluble mediators that will facilitate adhesion to the endothelium and transmigration.⁵ Human mast cells affect myofibroblast proliferation, collagen synthesis, and contractile activity, which influences scar formation in the later stages of the healing process,⁷ because myofibroblasts and fibroblasts are the major source of extracellular matrix.⁸

CD3⁺ T lymphocytes as well as cytotoxic populations of the immune system, like natural killer (NK) cells, are involved in the process of cutaneous wound healing in temporal sequences, which suggests that they may be involved in its modulation. NK cells are usually found in the initial stages of healing, among inflammatory cells,⁹ as well as B lymphocytes, which are found throughout the inflammatory process.^{10,11} Antibody-producing lymphocytes are variably affected by injury, probably following the alterations in T lymphocyte function, as a result of their close interaction with helper T cells.⁵

The establishment of an inflammatory reaction involving cell migration and the release of arachidonic acid mediators are crucial steps at the beginning of the wound healing process and final tissue repair. In addition to membrane phospholipid synthesis and intracellular signaling processes that control cell division and proliferation,¹² the unsaturated fatty acids, like linolenic and linoleic acids, are important arachidonic acid precursors and may be able to modify the inflammatory response and, consequently, the healing processes.

Fatty acids, in the form of phospholipids, are fundamental constituents of plasma membranes. These components are particularly important in leukocyte membranes, exerting major influences in the immune response.¹³ Among the fatty acids present in plasma membranes there are those which are polyunsaturated fatty acids (PUFA) which, in addition to their structural role, can modulate cell-cell interaction and intracellular signal transduction.¹³ N-3 and n-6 PUFA are capable of stimulating epithelial cell proliferation *in vitro*,¹⁴ a fundamental role during wound healing. PUFAs are also the primary precursors of many lipoic mediators with crucial functions in the inflammatory process,¹⁵ like vascular contraction, chemotaxis, adhesion, transmigration and cellular activation.¹⁶⁻¹⁸ Arachidonic acid, an n-6 PUFA, and its metabolites are mediators of several events during wound healing, such as cellular growth, angiogenesis, and extracellular matrix synthesis.¹⁹

Prostanoids, like prostaglandins, are involved both in the initial stage of tissue repair and during cellular spread and migration,¹⁹ and the utilization of anti-inflammatory drugs is an important factor that can influence the evolution of the healing process.²⁰

To examine the effects of topically administered agents on epidermal metabolism and on wound healing, various *in vitro* and *in vivo* models exist.²¹ Knowledge of the regulation of the proliferative tissue responses may allow the development of treatment regimes that can generate better conditions for tissue homeostasis and optimize the healing process.²¹ It has been shown that in experiments using topical administration of codfish oil that is rich in PUFA, animals had a faster

healing process as measured by reduction of the wound area.²¹ Conversely, dietary supplementation with an n-3 PUFA retarded re-epithelialization of surgical wounds in dogs.²² PUFA n-3 and n-6, in addition to their participation in the biosynthesis of inflammatory mediators, are also substrates, together with n-9 monounsaturated fatty acids, for the synthesis of membrane phospholipids, contributing to the control of signaling mechanisms of cell proliferation.¹²

The aim of the current studies was to evaluate the influence of topically administered essential (linolenic and linoleic acids) and nonessential (oleic acid) unsaturated fatty acids on the healing of mouse cutaneous wounds by means of immunofluorescence, macroscopic and microscopic morphometry and local nitric oxide detection.

MATERIALS AND METHODS

Linolenic (n-3), linoleic (n-6), and oleic (n-9) acids (Sigma Chemical Co., St. Louis, MO) were prepared at 30 μ M in a solution of glycerol and 0.02 M Tris-HCl, pH 7.4 (1 : 1 in volume), at the moment of use, to avoid oxidation.

Animal procedures

Ninety-six male BALB/c mice 6–8 weeks old were housed in individual cages with water and food *ad libitum*, in 12-hour dark-light cycles. All experimental procedures followed the guidelines of the Brazilian Council for Use of Animals in Research. Mice were anesthetized with intraperitoneal injection of 125 mg/kg ketamine (Happyvet-Pharma, Buenos Aires, Argentina) plus 12.5 mg/kg xylazine (Virbac do Brazil, São Paulo, Brazil), according to Demoyer et al.,²³ followed by shaving of the skin at the wounding site. After shaving, the skin was cleaned with a sterile water-soaked gauze and covered with a sterile surgical cloth, after which an elliptical area of approximately 20 mm² of skin was surgically removed from the dorsal region of the animals. The wound was immediately treated with topical application of 30 μ M of each fatty acid n-3 (Group I), n-6 (Group II), n-9 (Group III), or vehicle (Group IV), in a volume of 50 μ l. Treatment was repeated once daily for 20 days. No cleansing was done before wound treatment to avoid interference of antiseptic agents in wound closure. Three animals from each group were euthanized at time points 15 minutes, 3 hours, 24 hours, 48 hours, 5 days, 10 days, 15 days, and 20 days postsurgery. Tissue samples were collected and processed for NO detection, microscopic morphometry, and immunofluorescence.

Sample processing

The wound and surrounding skin were removed at each time point and divided into three parts: one was flash frozen in liquid nitrogen for NO production assay; the second part was embedded into preservative freezing medium (Tissue-Tek OCT Compound, Miles Inc., Elkhart, IN) for immunofluorescence staining; and the third part was fixed in 10 percent formalin and embedded in glycol-methacrylate resin (HistoResin, Reichert-Jung, Heidelberg, Germany) for microscopic morphometric analysis.

Macroscopic analysis of wound closure

To evaluate the wound closure under the different treatment regimens, the wounds were measured daily (with the exception of the 15-minute and 3-hour intervals) with the aid of a caliper (Vernier Caliper, Mitutoyo, Japan). The largest and smallest diameters of the wounds were measured from the edges of the original skin incision and the area (S) was calculated as $S = \pi ab$, where a and b correspond to one-half of the largest and one-half of the smallest diameter, respectively. All measurements were done directly on the animals, by the same examiner. Wound closure was defined as a reduction of the wound area and results were expressed as the mean $\pm$ standard deviation of wound area of all animals in each group.

NO assay

Preparation of wound lysates and nitrite determination were done according to Frank et al.²⁴ with modifications. Briefly, flash-frozen tissue samples were thawed in lysis buffer consisting of 0.02 M Tris-HCl, pH 7.4, 1 percent Triton-X-100 (Sigma), 10 percent glycerol, 500 mM ethylenediaminetetraacetic acid (EDTA, Sigma) and 1.6 mM phenylmethylsulfonyl fluoride (PMSF, Sigma). The tissue was homogenized and centrifuged at 20,000 $\times g$ for 2 hours at 4 °C, after which the supernatant was transferred to a fresh tube and mixed with 50 percent trichloroacetic acid (Synth, São Paulo, Brazil) in deionized water. After precipitation, the samples were centrifuged again for 1 minute (10,000 $\times g$, 4 °C) and the supernatant was collected for NO detection assay by the Griess method,²⁵ as follows. The reaction was done in duplicate wells by mixing 50 μ l of each sample with 50 μ l of Griess reagent (0.1 percent naphthyl-ethylenediamine [NEED] and 1 percent sulfanilamide in 2.5 percent phosphoric acid). After a 10-minute incubation at room temperature, the plates were read at 570 nm (Titertek Multiskan Plus, Lugano, Switzerland) and NO concentration was calculated using a standard curve ranging from 0.2 to 200 μ M of sodium nitrite.

Microscopic morphometric analysis

Tissue samples embedded in glycol-methacrylate resin were cut into 1.5- μ m-thick sections and stained by the PAS method for carbohydrates²⁶ or with a trichrome stain (acid fuchsin, Biebrich Scarlet and aniline blue) specific for connective tissue fibers (CTF). Sections were then examined under a microscope connected to a video camera and an image analysis system (HL-70/Image 97, Western Vision Software, Salt Lake City, UT). For each collected sample, we analyzed three fields of 6,300 μ m² to determine the area occupied by CTF and eight fields of 3,200 μ m² to count inflammatory cells (mast cells, eosinophils, and neutrophils) and blood vessels in the wound area.

Immunofluorescence procedure

To determine the presence of NK cells, activated B lymphocytes and cells expressing MHC class II in the inflammatory infiltrate, we performed an indirect immunofluorescence assay utilizing monoclonal antibodies against mouse NK1.1, B220, and I-A^d. These monoclonal antibodies were obtained as culture supernatants from clones PK136, RC3-2C2, and MKD6, respectively (American Type Culture Collection, Manassas, VA). We also used an irrelevant antibody as a negative control. Spleen sections were stained as positive control. Briefly, 4- μ m-thick frozen sections were fixed in acetone for 10 minutes at room temperature, followed by an incubation with 10 percent normal goat serum in 0.15 M phosphate buffered saline solution (PBS, pH 7.2) containing 0.1 percent glycine for 30 minutes at 37 °C. Tissue sections were then incubated with the undiluted primary antibodies for 45 minutes at 37 °C, followed by two washes of 5 minutes each in PBS and incubation with fluorescein isothiocyanate (FITC)-labeled secondary antibody (Sigma) for another 45 minutes at 37 °C. After two washes in PBS, slides were mounted and then analyzed with a Nikon epifluorescence microscope. Stained cells were counted under 1,000 $\times$ magnification.

Statistical analysis

To evaluate differences among the different groups in area occupied by CTFs, number of inflammatory cells, NO detection, differences in wound sizes, and time for wound closure, we performed ANOVA and Tukey tests. Significant results were considered when $p < 0.05$ (*) or $p < 0.01$ (**).

RESULTS

To evaluate whether the treatment with n-3, n-6, or n-9 fatty acids influenced the time for wound closure, daily measurements were taken from all animals. We observed a significant improvement in wound closure in the n-6 fatty acid treated group at 48 hours ($p < 0.05$,

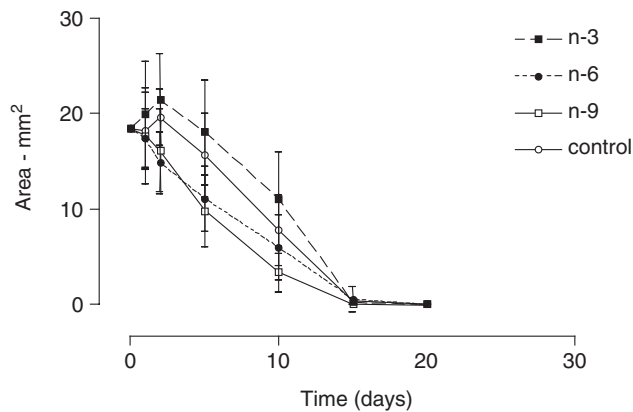


FIGURE 1. Measurement of wound area during the experimental period of treatment with n-3 (group I), n-6 (group II), n-9 fatty (group III) acids, and controls (group IV). While n-9 fatty acid-treated wounds maintained a trend to smaller wound areas, being significant at 5 days postsurgery, n-3 fatty acid-treated wounds had significantly larger wound areas when compared to n-9- and n-6-fatty acid-treated wounds, being the last group to completely close. Wounds treated with n-6 fatty acid were significantly smaller at 48 hours postsurgery, maintaining a trend to reduced areas until the end of the experiment. Results were considered statistically significant when $p < 0.05$ or $p < 0.01$.

Figures 1 and 2i) when compared to control-treated animals (Figure 2u). After 5 days of treatment, n-9 fatty acid-treated animals (Figure 2p) had the smallest wound area ($p < 0.05$, Figure 1) among the experimental groups, even when compared to control wounds (Figure 2v). A trend to smaller wound areas with n-6 and n-9 fatty acid treatments was observed throughout the first 10 days postsurgery in the present experiment (Figure 2g–k, m–q, respectively). Conversely, treatment with n-3 fatty acid did not significantly affect wound closure at 5 days postsurgery when compared to controls (Figure 2d, v, respectively). Nonetheless, at 5 and 10 days postsurgery, n-3 fatty acid-treated wounds were significantly larger than the n-9 fatty acid-treated wounds ($p < 0.05$ and $p < 0.001$, respectively; Figure 2d vs. 2p and 2e vs. 2q, respectively) and n-6 fatty acid treated animals ($p < 0.05$ for 5 and 10 days; Figure 2d vs. 2j and 2e vs. 2k, respectively). In addition to the wound sizes, we also observed the macroscopic characteristics of the wound, which revealed that wounds treated with n-9 fatty acid followed by n-6 fatty acid presented less edema at 48 hours when compared to control (Figure 2o vs. 2i vs. 2u, arrows). At 5 days postsurgery, while n-3 fatty acid-treated wounds had pronounced

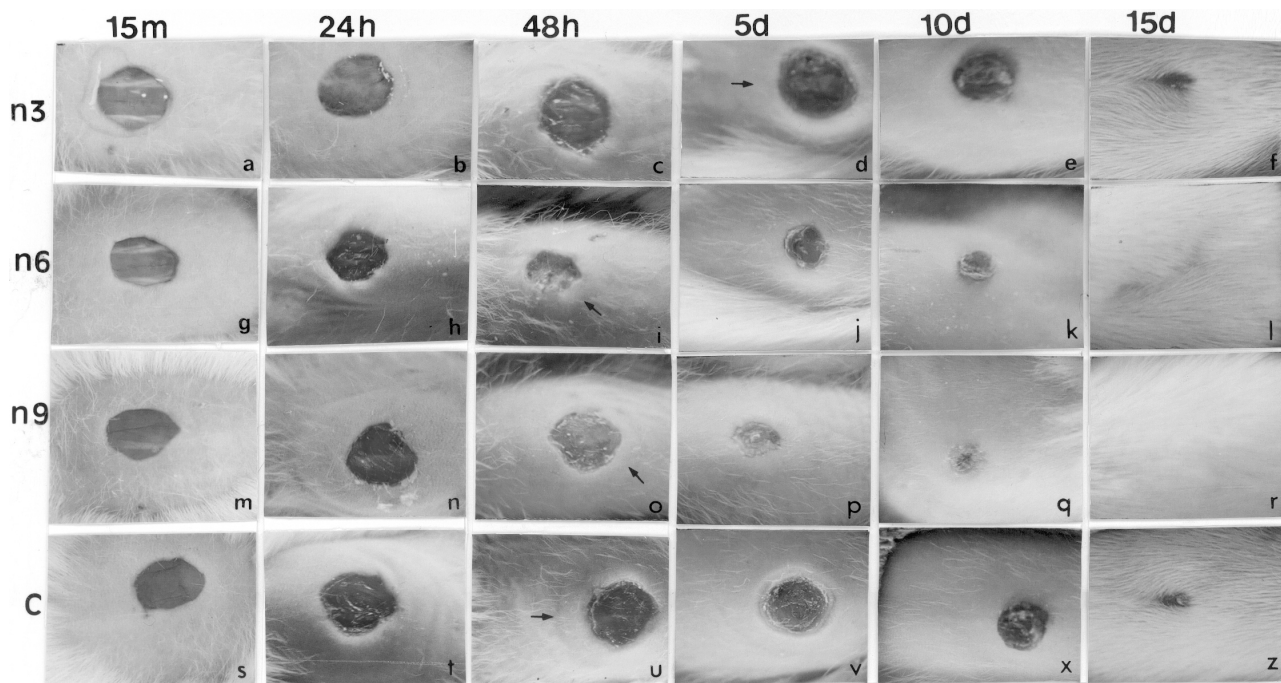


FIGURE 2. Macroscopic wound closure in treatment and control groups at different time points. In n-3 fatty acid-treated animals (a–f), there was little regression in wound size up to 10 days of treatment. Note the presence of a thicker fibrin clot and edema (arrow) surrounding the wound at day 5 (d). In the n-6 fatty acid group (g–l), edema surrounding the wound is clearly visible up to 48 hours (h–i, arrow points to edema), as well as a thin fibrin clot (l). In the n-9 fatty acid-treated animals (m–r), there is a small amount of edema around the wound at 48 hours (o, arrow) and visible regression in the wound area at 5 days (p) with an overall reduced edema and fibrin clot formation. In control-treated animals (s–z), edema surrounding the wound is visible up to 48 hours postsurgery (u, arrow), after which time there is formation of a fibrin clot that remained until the complete closure of the wounds. a, g, m, and s, 15 minutes postsurgery; b, h, n, and t, 24 hours postsurgery; c, i, o, and u, 48 hours postsurgery; d, j, p, and v, 5 days after surgery; e, k, q, and x, 10 days after surgery; f, l, r, and z, 15 days after surgery. Photographs were taken from a representative animal of each group. (Original magnification $\times 1.8$)

edema (Figure 2d, arrow) and a thicker fibrin clot cover, n-9 fatty acid-treated wounds had no visible edema and a thinner fibrin clot cover (Figure 2p). After 15 days of treatment, we could observe the complete wound closure in animals treated with the monounsaturated fatty acid n-9 (Figure 2r), followed by n-6 (Figure 2l). Control wounds closed on the 16th day and the n-3 fatty acid-treated group on the 17th day postsurgery (Figure 1).

Nitric oxide detection in the wound

To evaluate the effect of n-3, n-6, or n-9 fatty acid treatment on NO production at the site of wounds, we performed nitrite/nitrate measurements by the Griess method. Wound samples treated with n-6 fatty acid presented higher NO production when compared to control 15 minutes after surgery ($p < 0.05$, Figure 3), with a peak at 48 hours. On the other hand, n-9 fatty acid treatment significantly inhibited NO production up to 3 hours postsurgery ($p < 0.001$, Figure 3), being first detected 24 hours after surgery. In n-3 fatty acid-treated wounds, the peak of NO production happened at 3 hours following surgery and treatment, decreasing gradually to low levels at 48 hours (Figure 3).

Microscopic morphometric analysis

To determine whether treatment with n-3, n-6, or n-9 fatty acids influenced the number of inflammatory cells, blood vessels, and the amount of CTFs deposited at the site of the wound, we performed morphometric analysis, measuring the area occupied by CTFs as a percentage of the total area, as well as determining

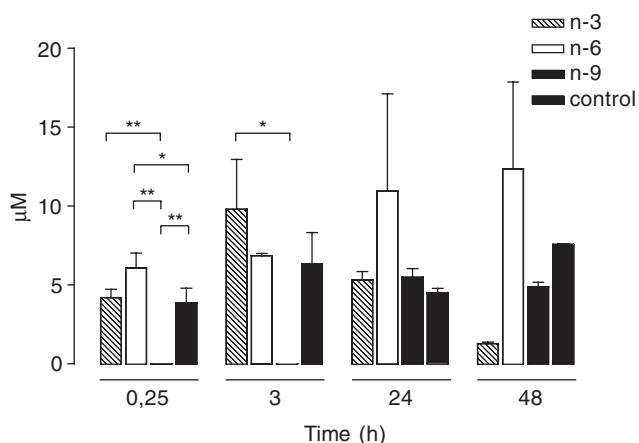


FIGURE 3. Local NO release in skin wounds treated topically with n-3 (group I), n-6 (group II), n-9 fatty (group III) acids and controls (group IV). After 15 minutes of surgery and treatment, NO levels were significantly higher in n-6 fatty acid-treated animals than in controls, peaking at 48 hours postsurgery. In wounds treated with n-3 fatty acid, NO peaked at 3 hours postsurgery, after which there was gradual reduction in NO release. Conversely, n-9 fatty acid seemed to inhibit NO release in the first hours after surgery, being first detected at 24 hours after treatment, with a slight decrease already at 48 hours. Values in brackets represent statistic differences: * $p < 0.05$; ** $p < 0.01$.

the number of neutrophils, eosinophils, mast cells, and blood vessels in the wound area. We observed a significant increase in CTF deposition in n-3 fatty acid-treated animals after 5 days of treatment, when compared to control ($p < 0.01$, Figures 4 and 5a vs. 5g), n-9 fatty acid ($p < 0.01$, Figures 4 and 5e vs. 5g), and n-6 fatty acid ($p < 0.05$, Figures 4 and 5c vs. 5g)-treated animals. At 10 days of treatment, n-3 fatty acid-treated animals still had significantly larger areas occupied by CTFs than n-9 fatty acid-treated animals ($p < 0.05$, Figures 4 and 5b vs. 5f). No significant differences were observed in the number of cells or blood vessels among the different groups at the time points tested (Table 1).

Immunofluorescent detection of immune cells

When we stained sections of the wounds with antibodies against NK cells, B lymphocytes, and I-A^{d+} cells, we found no significant differences among the groups (Table 2).

DISCUSSION

Repeated studies showing that unsaturated fatty acids can modify the production and activity of various components of the immune system have left unexplained the mode of action by which these compounds exert their effects. Several mechanisms have been proposed, including membrane fluidity,²⁷ lipid peroxidation,²⁸ prostaglandin production,²⁹ and regulation of gene expression.³⁰ Both n-3 and n-6 PUFA can alter the composition and function of membrane rafts through eicosanoid-independent mechanisms.³¹ Fatty acids may

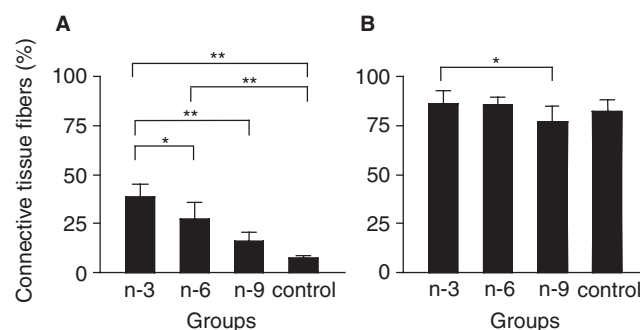


FIGURE 4. Area occupied by CTFs in wounds treated topically with n-3 (group I), n-6 (group II), n-9 fatty (group III) acids and controls (group IV). (A) Five days after surgery, wounds treated with n-3 fatty acid showed significantly larger area occupied by CTFs when compared to controls and n-6- and n-9 fatty acid-treated wounds. Moreover, wounds treated with n-6 fatty acid also presented larger areas occupied by CTFs when compared to controls, which was not observed for treatment with n-9 fatty acid. (B) Ten days of treatment; n-3 fatty acid-treated wounds still had a significantly larger area occupied by CTFs when compared to n-9 fatty acid treatment. Values in brackets represent statistic differences: * $p < 0.05$; ** $p < 0.01$.

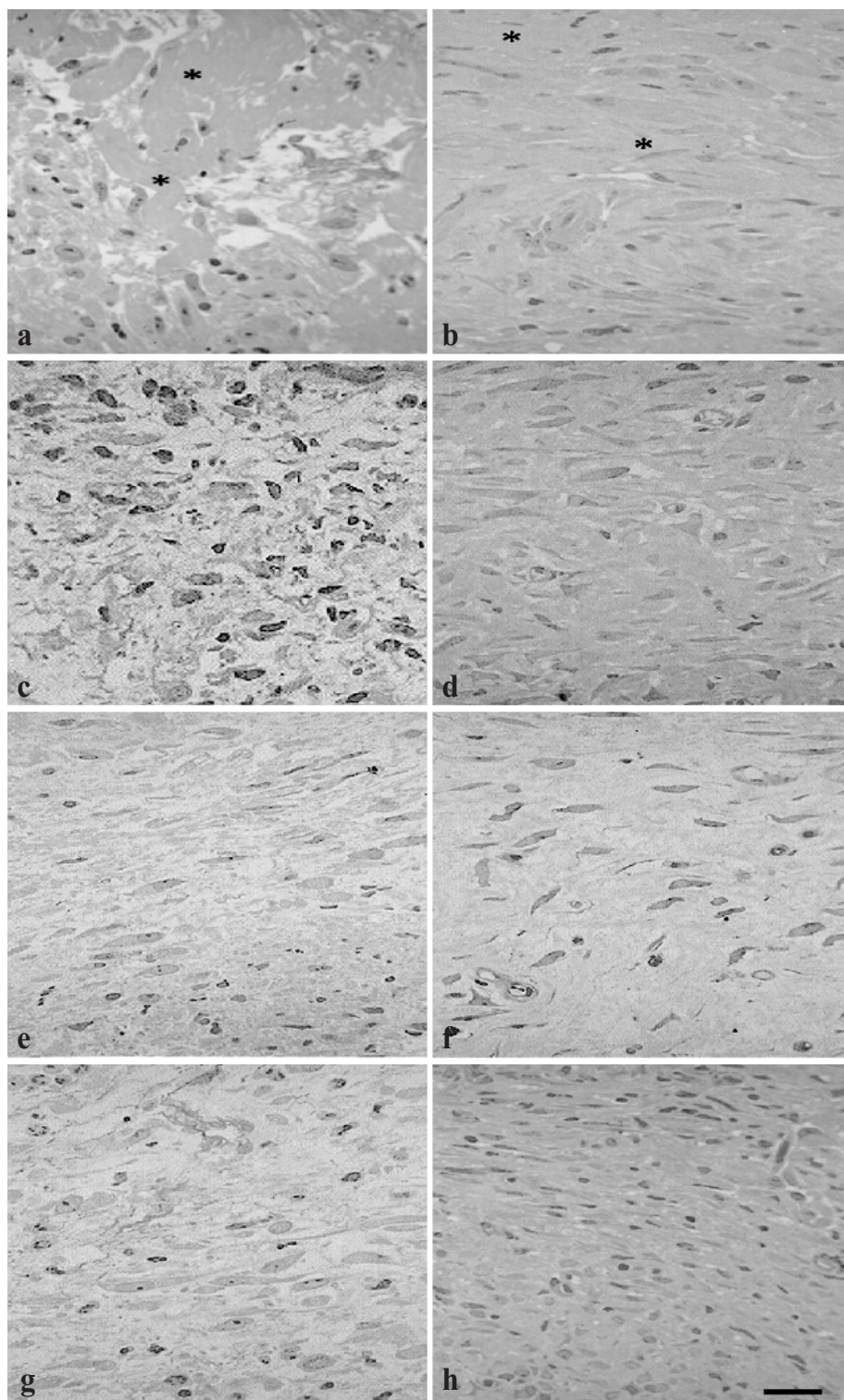


FIGURE 5. Photomicrographs of connective tissue subjacent to the wound. (a) and (b) Animals treated with n-3 fatty acid for 5 and 10 days, respectively. Note the extensive deposition of CTFs (*) in the n-3 fatty acid-treated group, with abundant collagen fibers especially at 5 days of treatment (a), when compared to (c) n-6 fatty acid, (e) n-9 fatty acid, and (g) control. (c) and (d) Animals treated with n-6 fatty acid for 5 and 10 days, respectively. The inflammatory infiltrate decreased by 10 days; however, fibroblasts and the area occupied by deposition of CTFs increased (d). (e) and (f) Animals treated with n-9 fatty acid for 5 and 10 days, respectively. Few inflammatory cells and CTFs are seen at both time points. (g) and (h) Control-treated animals after 5 and 10 days, respectively, showing several fibroblast and deposition of CTFs at day 5 (g) and a more compact matrix at day 10. Trichrome staining. Bar: 17 μ m.

Table 1. Number of blood vessels and inflammatory cells per 25,600 μm^2 of wound area at different times after wounding

Time	Blood vessels*				Neutrophils				Eosinophils				Mast cells			
	n-3	n-6	n-9	control	n-3	n-6	n-9	control	n-3	n-6	n-9	control	n-3	n-6	n-9	control
15 minutes	11 $\pm$ 4	10 $\pm$ 2	3 $\pm$ 2	7 $\pm$ 3	15 $\pm$ 5	8 $\pm$ 5	10 $\pm$ 5	4 $\pm$ 2	1 $\pm$ 1	2 $\pm$ 0	3 $\pm$ 1	4 $\pm$ 2	3 $\pm$ 2	5 $\pm$ 1	3 $\pm$ 1	3 $\pm$ 2
3 hours	4 $\pm$ 3	6 $\pm$ 3	5 $\pm$ 2	8 $\pm$ 4	2 $\pm$ 2	6 $\pm$ 2	2 $\pm$ 2	8 $\pm$ 9	3 $\pm$ 2	10 $\pm$ 2	2 $\pm$ 3	6 $\pm$ 8	3 $\pm$ 2	5 $\pm$ 3	4 $\pm$ 2	4 $\pm$ 1
24 hours	3 $\pm$ 2	3 $\pm$ 5	2 $\pm$ 2	0 $\pm$ 0	30 $\pm$ 19	19 $\pm$ 8	16 $\pm$ 5	12 $\pm$ 7	12 $\pm$ 3	8 $\pm$ 9	14 $\pm$ 8	8 $\pm$ 6	1 $\pm$ 1	0 $\pm$ 0	1 $\pm$ 1	0 $\pm$ 0
48 hours	4 $\pm$ 2	4 $\pm$ 2	2 $\pm$ 2	1 $\pm$ 1	20 $\pm$ 15	31 $\pm$ 10	34 $\pm$ 18	31 $\pm$ 1	5 $\pm$ 4	20 $\pm$ 7	12 $\pm$ 4	12 $\pm$ 12	1 $\pm$ 1	2 $\pm$ 2	0 $\pm$ 0	0 $\pm$ 0
5 days	2 $\pm$ 2	1 $\pm$ 1	1 $\pm$ 1	1 $\pm$ 1	15 $\pm$ 7	19 $\pm$ 3	31 $\pm$ 25	16 $\pm$ 8	10 $\pm$ 14	6 $\pm$ 2	24 $\pm$ 4	11 $\pm$ 8	1 $\pm$ 1	1 $\pm$ 1	0 $\pm$ 0	0 $\pm$ 0
10 days	3 $\pm$ 1	6 $\pm$ 4	1 $\pm$ 1	4 $\pm$ 2	1 $\pm$ 1	4 $\pm$ 2	2 $\pm$ 1	15 $\pm$ 9	0 $\pm$ 0	1 $\pm$ 1	4 $\pm$ 2	5 $\pm$ 2	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	1 $\pm$ 1

* Results are expressed as the mean $\pm$ standard deviation (SD) of the counts done in three animals of each group.

also generate other lipid mediators such as intermediate hydroperoxides, with anti-inflammatory activities, or lipoxins, which can alter the immune response toward a Th2 profile³² and thereby alter tissue repair.

The presence of a higher number of unsaturations makes a fatty acid more likely to be oxidized and consequently leads to a delay in wound healing. In agreement with this, it has been shown that inhibition of lipid peroxidation diminishes the delay effect on tissue repair,³³ while incorporation of high amounts of oleic acid, a monounsaturated fatty acid, protects against lipid peroxidation.^{34,35}

In this study, we analyzed the process of skin wound healing, as measured by total wound closure, during topical administration of n-3 and n-6 PUFA and n-9 monounsaturated fatty acid. Wound closure results from a combination of contraction of the wound, mediated by myofibroblasts, and re-epithelialization, which reflects the advancement of the epithelium over the granulation tissue.²¹ Our results indicated macroscopic differences in wound closure among the treatment groups only in the initial response phase, suggesting a specific effect of n-6 PUFA on wound closure in the first 48 hours after surgery and a tendency to a delay in wound closure in the first 10 days after surgery in the group treated with n-3 PUFA. The group receiving the monounsaturated n-9 fatty acid showed a tendency toward faster wound closure in the first 10 days of treatment. These results disagree, in part, with those obtained in a study where the authors observed faster wound repair after topical administration of codfish oil,²¹ rich in PUFA. This difference may be due to the fact that codfish oil has

several components that may influence healing, while we used purified preparations. On the other hand, the results of our study agree with other findings²² that showed that animals given a diet rich in n-3 PUFA had a delay in total wound closure, as a reflection of the decrease in both re-epithelialization and contraction of skin wounds. Also, animals treated with lysophosphatidic acid, a growth factor derived from phospholipids, showed faster skin wound closure without significant differences in the experimental inflammatory process.²³

We have also observed that the administration of PUFA and n-9 fatty acid altered the deposition of CTF in the wound site, such that the area occupied by these fibers was greater when treatment was done with the more unsaturated fatty acids. These results agree with a study that showed that eicosapentaenoic acid, an n-3 PUFA, stimulates collagen synthesis by fibroblasts after 72 hours in culture. Moreover, the same study showed that arachidonic acid, an n-6 PUFA, induces less collagen production, yet higher levels of prostaglandin E₂.³⁶

In the present work, we have also observed that mice treated with n-3 fatty acid had a larger area occupied by CTFs in the healing wound and slower wound closure in the first 10 days after surgery, while n-9 fatty acid-treated mice had one of the smallest areas occupied by connective tissue and faster wound closure in the same period. Although we did not measure wound contraction and epithelialization separately, these results suggest that n-9 fatty acid treatment may have favored epithelialization over wound contraction, as we did not observe larger areas of CTFs deposited in

Table 2. Number of NK cells, B lymphocytes and I-A^{d+} cells in the wound area§, at different time points after wounding

Time	I-A ^{d+} cells*				B lymphocytes				NK cells			
	n-3	n-6	n-9	control	n-3	n-4	n-9	control	n-3	n-4	n-9	control
15 minutes	2 $\pm$ 1	3 $\pm$ 1	1 $\pm$ 0	0 $\pm$ 0	1 $\pm$ 1	0 $\pm$ 0	1 $\pm$ 0	1 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0
3 hours	1 $\pm$ 1	2 $\pm$ 2	0 $\pm$ 0	0 $\pm$ 0	1 $\pm$ 1	1 $\pm$ 1	0 $\pm$ 0	1 $\pm$ 1	1 $\pm$ 1	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0
24 hours	1 $\pm$ 1	2 $\pm$ 1	0 $\pm$ 0	0 $\pm$ 0	1 $\pm$ 0	1 $\pm$ 1	0 $\pm$ 0	1 $\pm$ 1	1 $\pm$ 1	0 $\pm$ 0	1 $\pm$ 1	1 $\pm$ 1
48 hours	2 $\pm$ 1	2 $\pm$ 0	0 $\pm$ 0	2 $\pm$ 2	2 $\pm$ 1	2 $\pm$ 1	2 $\pm$ 1	1 $\pm$ 1	1 $\pm$ 0	2 $\pm$ 2	2 $\pm$ 0	1 $\pm$ 1

§ Analysis of eight fields with 1000 $\times$ magnification/slide/animal.

* Results were expressed as the mean $\pm$ standard deviation (SD) of the counts done in two animals of each group.

this group. In addition, the treatment with n-9 mono-unsaturated fatty acid may also have induced a less intense local inflammatory response and therefore faster wound closure in the first days after surgery. Once n-9 fatty acid competes with PUFA for insertion into membrane phospholipids, it is no longer a substrate for the oxidases that will produce lipid inflammatory mediators.³⁷

Several studies have shown a role for NO in the process of tissue repair.² In this study, n-9 fatty acid inhibited NO production in the first hours after surgery, which may also have contributed toward a faster re-epithelialization of the wounds. Davda and collaborators³⁸ have shown that oleic acid inhibits iNOS activity *in vivo*, suggesting that in our study, treatment with n-9 fatty acid may have inhibited iNOS, which is usually highly induced in skin lesions.² Conversely, we observed a tendency to accumulate neutrophils and eosinophils in the first 24 hours of n-3 fatty acid treatment, suggesting a more intense early inflammatory response in this group, which may have contributed to a delay in wound closure in the first days after surgery. This impaired closure could result from a delay in the resolution of the exudative phase of the inflammatory process, which is crucial for repair to occur.

Newly formed capillaries participate in the formation of granulation tissue and provide oxygen to the healing tissues.³⁹ In the present work, we did not find a significant difference in either the number of blood vessels or the number of inflammatory cells in the wound site. Nonetheless, the significant differences observed in CTF deposition, time of wound closure, and especially, in NO production suggest functional alterations in the cells at the inflammatory site after treatment with n-3, n-6, and n-9 fatty acids.

The results presented here therefore indicate a relevant role for n-3, n-6, and n-9 fatty acids in skin wound healing, which could lead to improvement of therapeutic resources in the treatment of skin wounds. Utilization of oleic acid could lead to better closure, particularly in cases where excessive collagen deposition might lead to an unsatisfactory aesthetic or functional results. In addition, it could play an important role in the treatment of open wounds, such as skin burns, where faster wound closure would be beneficial for the patient. On the other hand, in diabetic patients, who frequently present difficulties in wound healing, drug formulations containing linolenic acid could be relevant in stimulating higher CTF deposition and better repair. In this way, both n-3 and n-6 PUFA as well as monounsaturated n-9 fatty acids may represent important components to be considered in drug formulations for use during the processes of skin wound healing.

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